CSE 4631

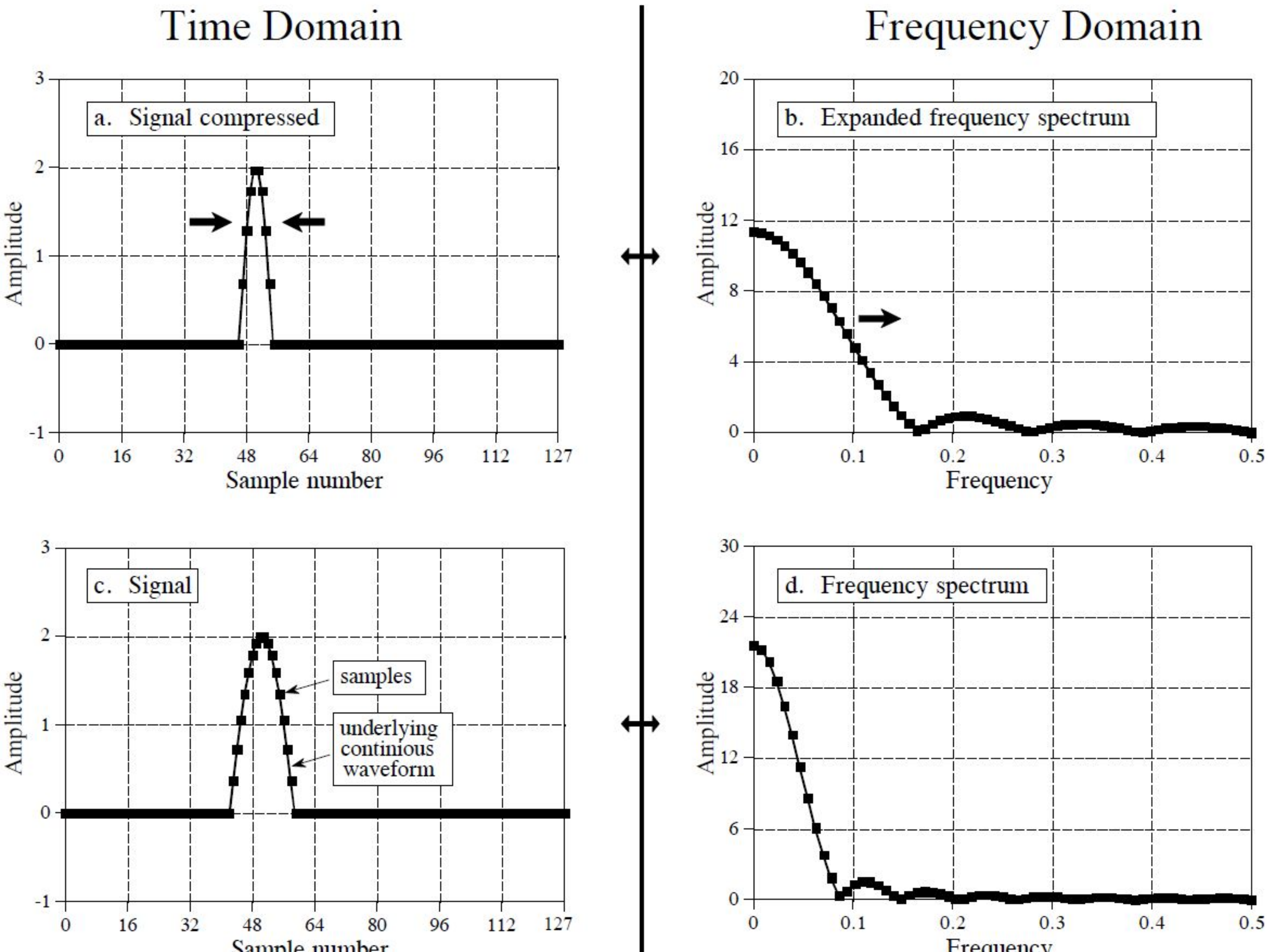
Digital Signal Processing

Week 13

Compression and Expansion, Multirate methods

- A compression of the signal in one domain results in an expansion in the other, and vice versa.
- If an event happens faster (compressed in time), that means it must be composed of higher frequencies. (more cycles per sample).
- If an event happens slower (expanded in time), it is composed of lower frequencies.
- **For continuous signals**, if $X(f)$ is the Fourier transform of $x(t)$, then $1/k \times X(f/k)$ is the Fourier transform of $x(kt)$.
- If we take it to the two extremes, this pattern holds –
  - If the time domain is compressed so far that it becomes an *impulse*, the corresponding frequency spectrum is expanded so far that it becomes a *constant value*.
  - And vice versa!

Compression and Expansion, Multirate methods (contd.)



Compression and Expansion, Multirate methods (contd.)

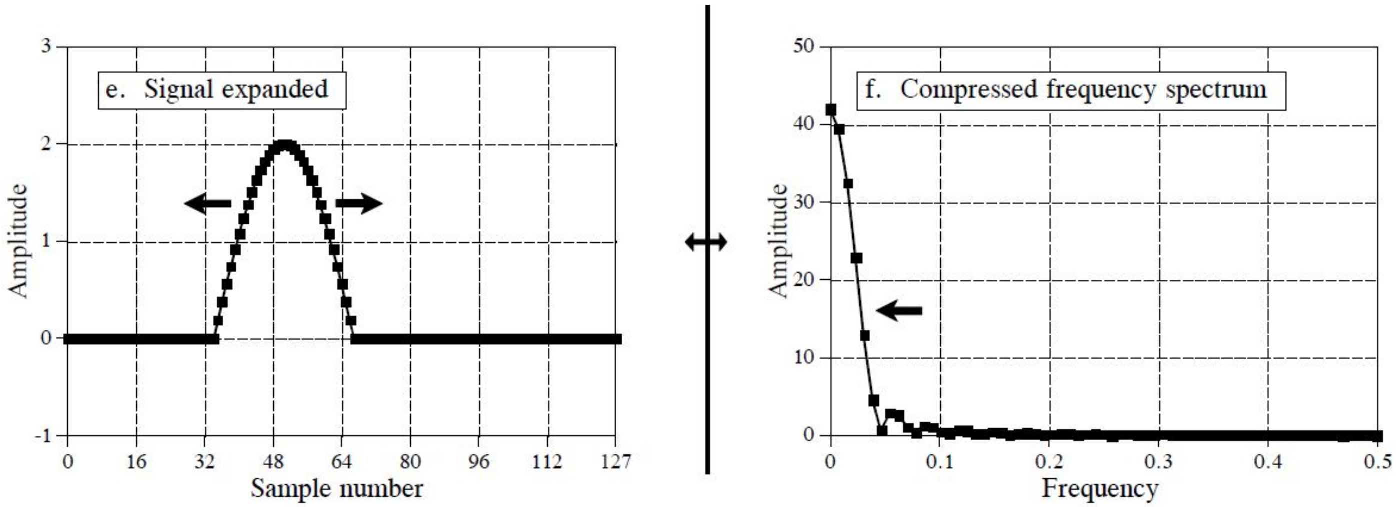


FIGURE 10-12
Compression and expansion. Compressing a signal in one domain results in the signal being expanded in the other domain, and vice versa. Figures (c) and (d) show a discrete signal and its spectrum, respectively. In (a) and (b), the time domain signal has been compressed, resulting in the frequency spectrum being expanded. Figures (e) and (f) show the opposite process. As shown in these figures, discrete signals are expanded or contracted by expanding or contracting the underlying continuous waveform. This underlying waveform is then resampled to find the new discrete signal.

Compression and Expansion, Multirate methods (contd.)

Discrete signals behave in a similar fashion, but there are some issues:

- The first issue is **aliasing**. If the time domain signal is compressed so far that several points in the frequency spectrum are pushed to frequencies beyond 0.5, the resulting aliasing breaks the simple expansion/contraction relationship. This type of aliasing can also happen in the time domain.
- A second issue is **to define exactly what it means to compress or expand a discrete signal**.
  - **Consideration 1**: First, the underlying continuous curve is compressed that the samples lie on, and then we resample the new continuous curve using the same sampling rate to find the new discrete signal.
  - **Consideration 2**: We kept the underlying continuous waveform the same, but resample at a different sampling rate – **Multirate techniques!**
- If more samples are added, it is called **interpolation**. If fewer samples are used to represent the signal, it is called **decimation**.

Compression and Expansion, Multirate methods (contd.)

Consideration 1: The sampling rate has been kept constant but the underlying waveform has been expanded to be eight times.

Consideration 2: The underlying waveform has been kept constant, but the sampling rate has been increates by a factor of eight – **interpolation**

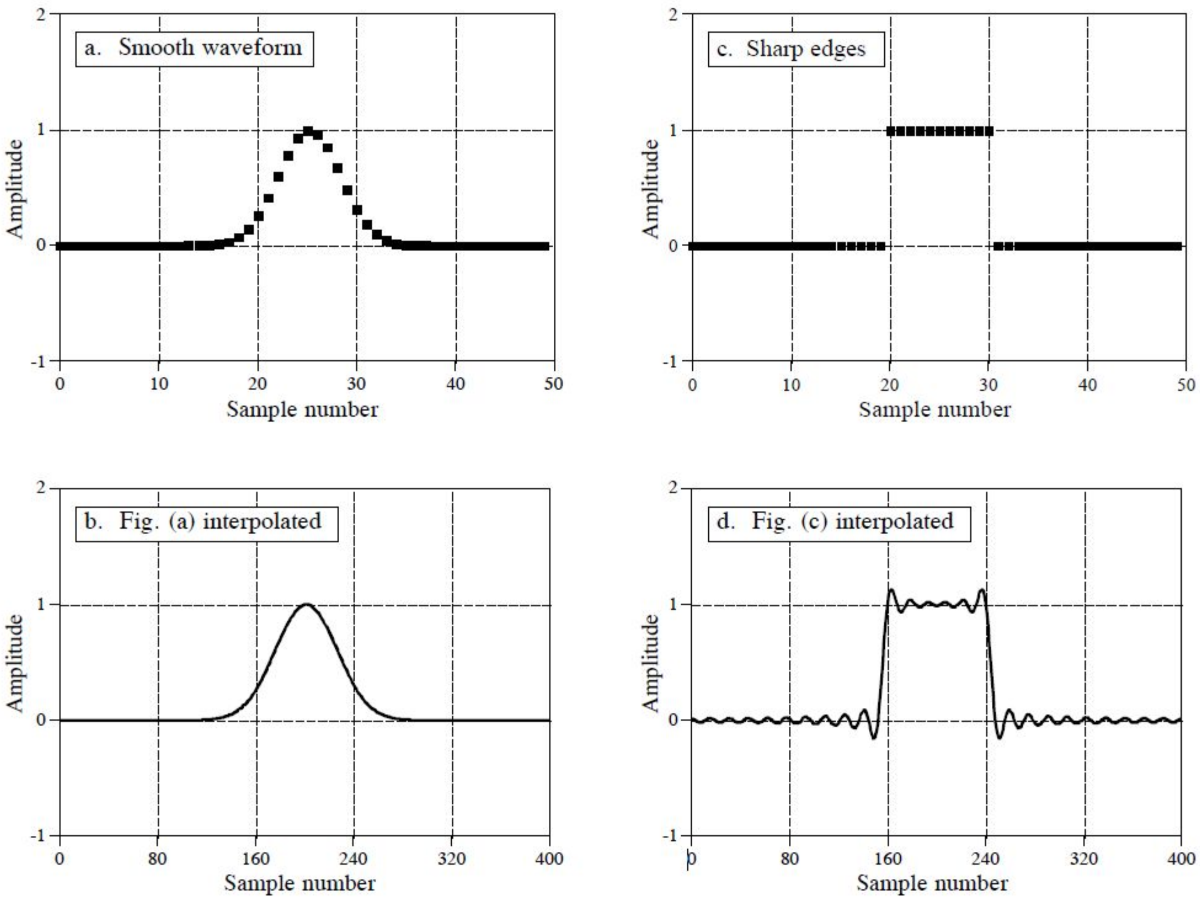


FIGURE 10-13
Interpolation by padding the frequency domain. Figures (a) and (c) each consist of 50 samples. These are interpolated to 400 samples by padding the frequency domain with zeros, resulting in (b) and (d), respectively. (Figures (b) and (d) are discrete signals, but are drawn as continuous lines because of the large number of samples).

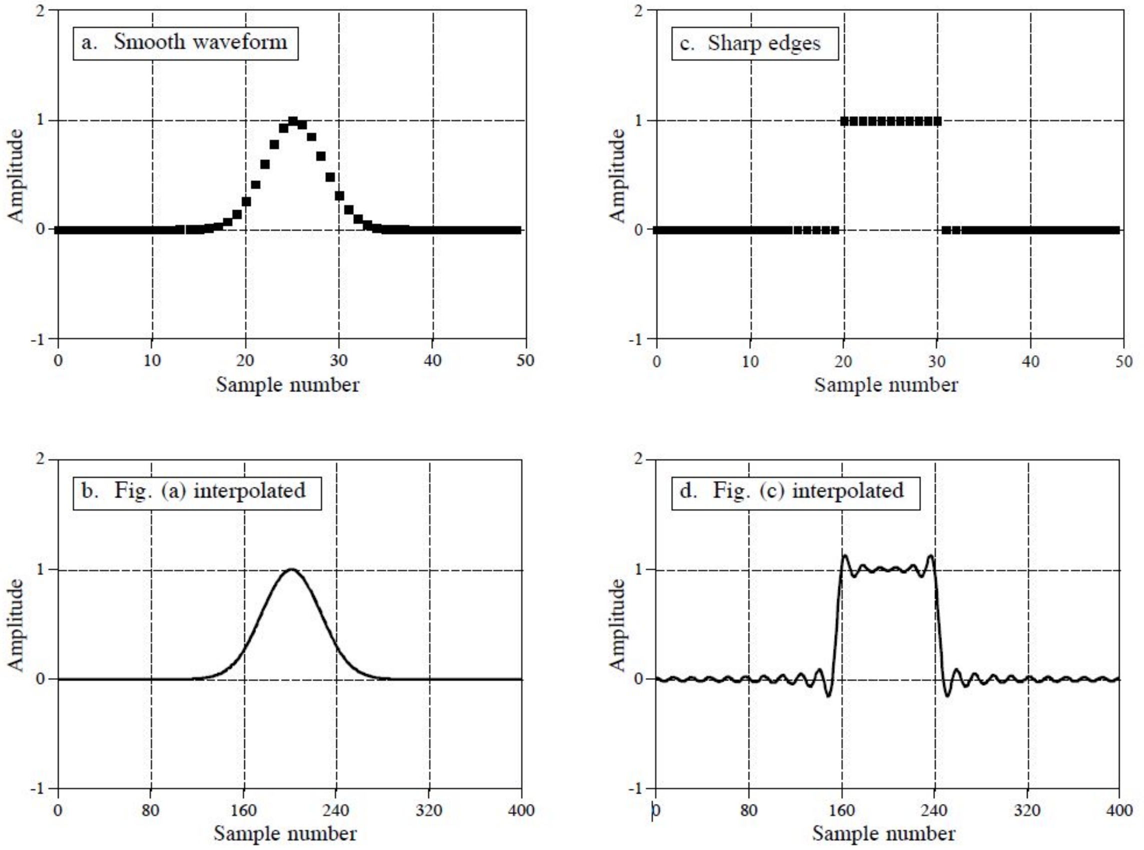
Compression and Expansion, Multirate methods (contd.)

How can you interpolate a time domain discrete signal?

- Remember that a finer sampling of a frequency spectrum (more samples between frequency 0 and 0.5) can be obtained by padding the time domain with zeros before taking the DFT.
- Duality allows this to work in the opposite direction! If we want a finer sampling in the time domain (interpolation), pad the frequency spectrum with zeros before taking the inverse DFT.
- **Example:** Say, we want to interpolate a 50 sample signal into a 400 sample signal. It's done like this:
  - Take the 50 samples and add zeros to make the signal 64 samples long.
  - Use a 64 point DFT to find the frequency spectrum, which will consist of a 33 point real part and a 33 point imaginary part.
  - Pad the right side of the frequency spectrum (both the real and imaginary parts) with 224 zeros to make the frequency spectrum 257 points long.
  - Use a 512 point Inverse DFT to transform the data back into the time domain. This will result in a 512 sample signal that is a high resolution version of the 64 sample signal.
- The first 400 samples of this signal are an interpolated version of the original 50 samples.

Compression and Expansion, Multirate methods (contd.)

In figure (d), the oscillatory behaviour arises at edges or other discontinuities in the signal. This also includes any discontinuity between sample 0 and N-1, since the time domain is viewed as being circular. This overshoot at discontinuities is called the **Gibbs effect**.



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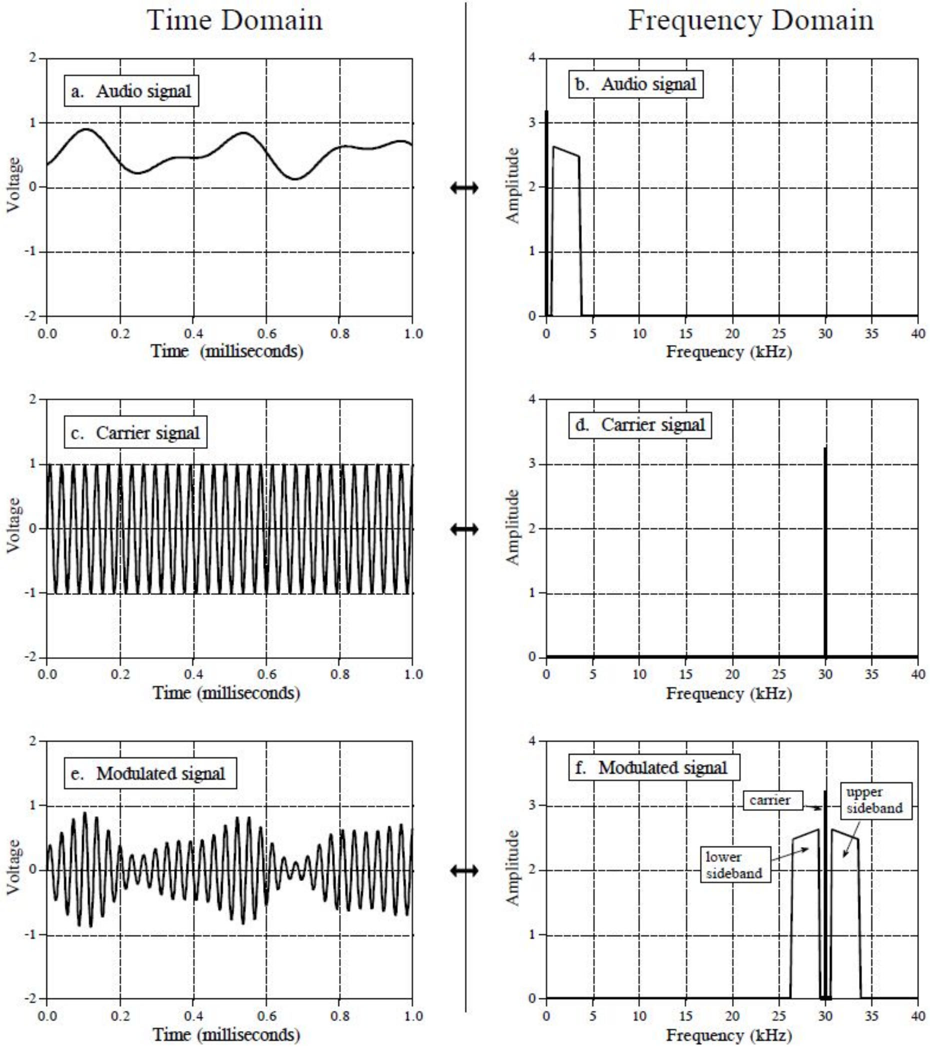
Multiplying Signals (Amplitude Modulation)

- Modulation is the process of merging two signals to form a third signal with desirable characteristics of both. This always involves nonlinear processes such as multiplication; you cannot just add two signals together.
- **Example:** In radio communication, modulation results in radio signals that can propagate long distances *and* carry along audio or other information.
- Among many modulation techniques developed in the radio communication discipline, one of the simplest is **amplitude modulation**.
- It is an example of the following **two scenarios**:
  - Multiplication in time domain corresponds to convolution in the frequency domain.
  - How the elusive negative frequencies enter into everyday science and engineering problems.

- Figure (a) shows an audio signal with a DC bias such that the signal always has a positive value.
- Figure (b) shows that its frequency spectrum is composed of frequencies from 300 Hz to 3 kHz, the range needed for voice communication, plus a spike for the DC component. All other frequencies have been removed by analog filtering.
- Figures (c) and (d) show the carrier wave, a pure sinusoid of much higher frequency than the audio signal.
- In the time domain, amplitude modulation consists of multiplying the audio signal by the carrier wave.
- As shown in (e), this results in an oscillatory waveform that has an instantaneous amplitude proportional to the original audio signal.
    - In DSP jargon, it is said that the *envelope* of the carrier wave is equal to the modulating signal.

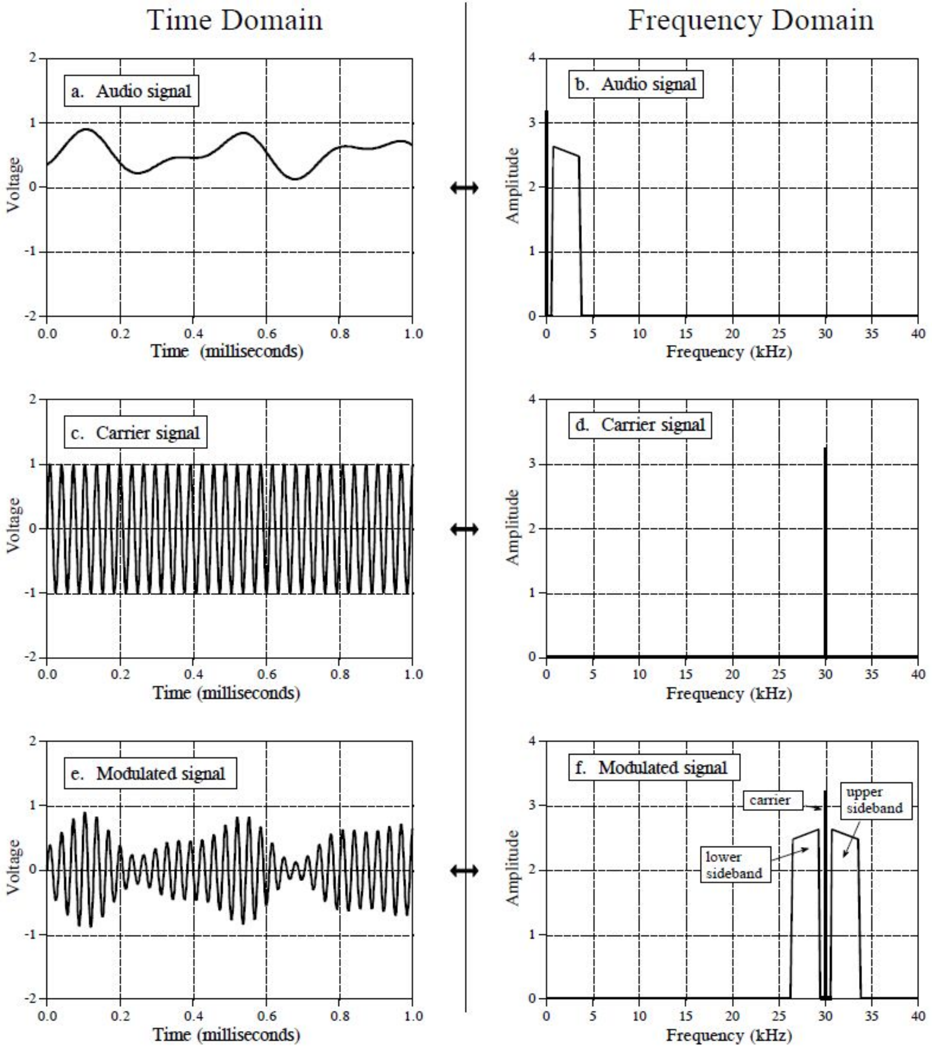


Since multiplication in the time domain corresponds to convolution in the frequency domain, the spectrum of the modulated signal is the spectrum of the audio signal shifted to the frequency of the carrier. (since the frequency spectrum of the carrier is a shifted delta function)

This results in **three components:**

1. **A carrier wave:** corresponds to the DC component of the original audio signal
2. **An upper sideband:** corresponds to the positive frequencies between 0.3 and 3 kHz
3. **A lower sideband:** corresponds to the negative frequencies between -0.3 and -3 kHz

Even though the negative frequencies in the original audio signal are somewhat elusive and abstract, the resulting frequencies in the lower sideband are real.



Fast Fourier Transform

- Based on the complex DFT, a more sophisticated version of the *real DFT*.
- **The real DFT**
  - Transforms an N point time domain signal into two $N/2+1$ point frequency domain signals.
  - The time domain signal is called just that: the time domain signal.
  - The two signals in the frequency domain are called the real part and the imaginary part, holding the amplitudes of the cosine waves and sine waves, respectively.
- **The complex DFT**
  - Transforms two N point time domain signals into two N point frequency domain signals.
  - The two time domain signals are called the *real part* and the *imaginary part*, just as are the frequency domain signals.

Fast Fourier Transform (contd.)

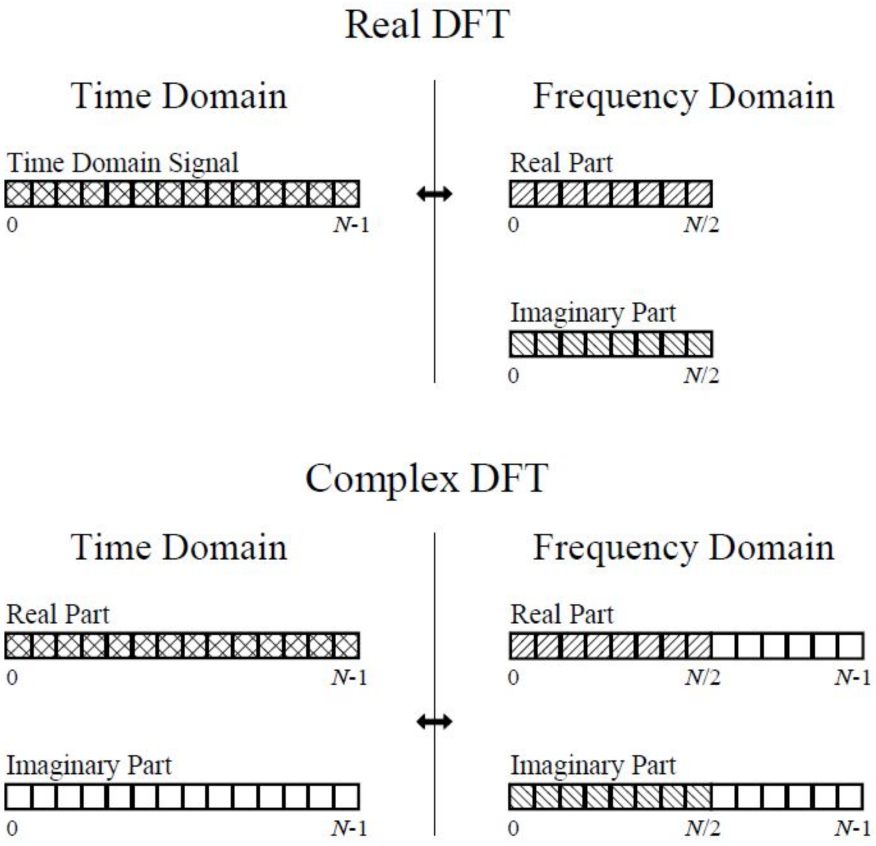


FIGURE 12-1
Comparing the real and complex DFTs. The real DFT takes an N point time domain signal and creates two $N/2 + 1$ point frequency domain signals. The complex DFT takes two N point time domain signals and creates two N point frequency domain signals. The crosshatched regions shows the values common to the two transforms.

Fast Fourier Transform

Calculation of the real DFT by means of the Complex DFT

- Suppose you have an N point signal.
- First, move the N point signal into the real part of the complex DFT's time domain, and then set all of the samples in the imaginary part to zero.
- Calculation of the complex DFT results in a real and an imaginary signal in the frequency domain, each composed of N points. Samples 0 through $N/2$ of these signals correspond to the real DFT's spectrum.
- Remember that the DFT's frequency domain is periodic when the negative frequencies are included.
- Therefore, one full period stretches from sample 0 to sample $N-1$, corresponding with 0 to 1.0 times the sampling rate.
- The positive frequencies sit between sample 0 and $N/2$, corresponding with 0 to 0.5.
- The other samples, between $N/2+1$ and $N-1$, contain the negative frequency values (which are usually ignored).

Fast Fourier Transform

How the FFT works

- First, decomposing an N point time domain signal into N time domain signals each composed of a single point.
  - Uses **interlaced decomposition** in theory.
  - Uses a **bit reversal sorting algorithm** in computer programs.
  - Requires Log_2N steps for this decomposition. (e.g., 4 steps for 16 point signal)
- The second step is to calculate the N frequency spectra corresponding to these N time domain signals.
  - The frequency spectrum of a 1 point signal is equal to itself!
  - That means nothing is required to do in this step
- Lastly, the N spectra are synthesized into a single frequency spectrum.

Fast Fourier Transform (contd.)

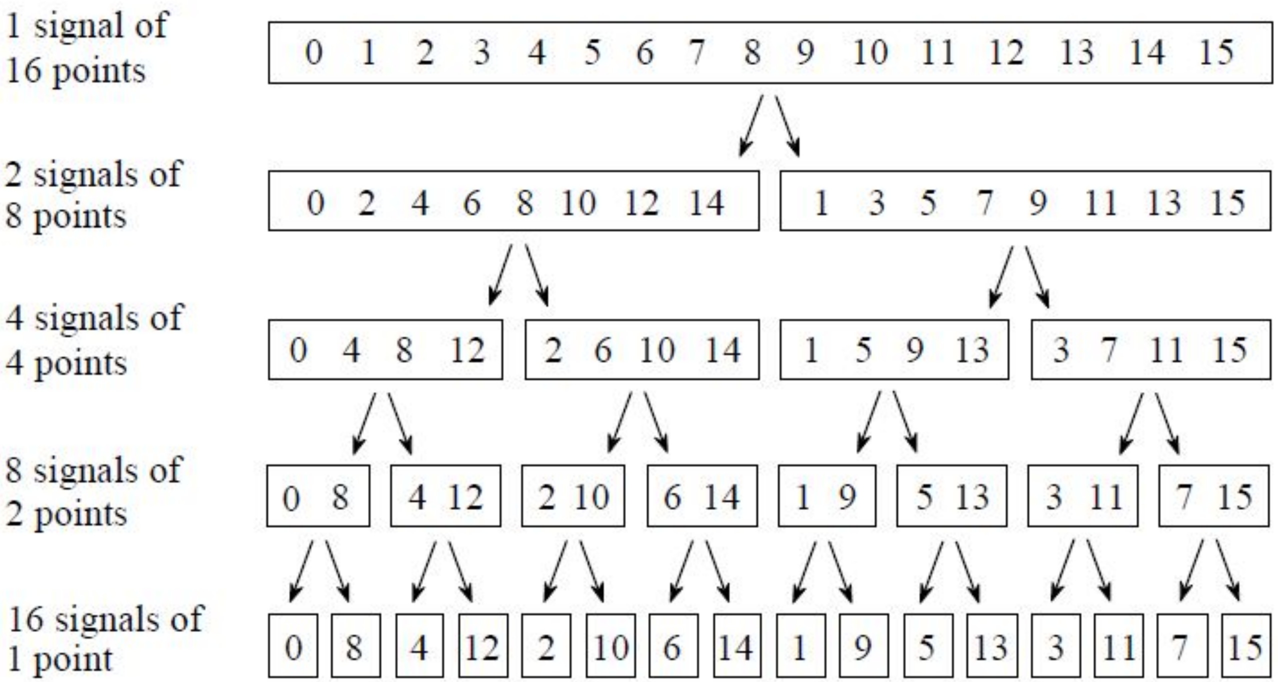


FIGURE 12-2
The FFT decomposition. An N point signal is decomposed into N signals each containing a single point. Each stage uses an *interlace decomposition*, separating the even and odd numbered samples.

Fast Fourier Transform (contd.)

Sample numbers in normal order

Decimal	*Binary*
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

Sample numbers after bit reversal

Decimal	*Binary*
0	0000
8	1000
4	0100
12	1100
2	0010
10	1010
6	0100
14	1110
1	0001
9	1001
5	0101
13	1101
3	0011
11	1011
7	0111
15	1111

FIGURE 12-3
The FFT bit reversal sorting. The FFT time domain decomposition can be implemented by sorting the samples according to bit reversed order.

Fast Fourier Transform

Combining the N frequency spectra

- This process has to be done in the exact reverse order that the time domain took place in. And the bit reversal shortcut is not applicable. We must go back one stage at a time.
    - In the first stage, 16 frequency spectra (1 point each) are synthesized into 8 frequency spectra (2 points each).
    - In the second stage, the 8 frequency spectra (2 points each) are synthesized into 4 frequency spectra (4 points each), and so on.
    - The last stage results in the output of the FFT, a 16 point frequency spectrum.
- This synthesis must undo the interlaced decomposition done in the time domain. For example,
    - Two frequency spectra, each composed of 4 points, are combined into a single frequency spectrum of 8 points.
    - The frequency domain operation must correspond to the time domain procedure of combining two 4 point signals by interlacing.

Fast Fourier Transform

Combining the N frequency spectra (contd.)

- Consider two time domain signals, $abcd$ and $efgh$. An 8 point time domain signal can be formed by two steps:
  - Dilute each 4 point signal with zeros to make it an 8 point signal. That is, $abcd$ becomes $a0b0c0d0$, and $efgh$ becomes $0e0f0g0h$.
  - Then add the signals together. So the resulting 8 point signal will be $aebfcgdh$.
- Diluting the time domain with zeros corresponds to a duplication of the frequency spectrum. Therefore, the frequency spectra are combined in the FFT by duplicating them, and then adding the duplicated spectra together.
- In order to match up when added, the two time domain signals are diluted with zeros in a slightly different way. In one signal, the odd points are zero, while in the other signal, the even points are zero. In other words, one of the time domain signals ($0e0f0g0h$) is shifted to the right by one sample.
- This time domain shift corresponds to multiplying the spectrum by a sinusoid.
- Remember that a shift in the time domain is equivalent to convolving the signal with a shifted delta function. This multiplies the signal's spectrum with the spectrum of the shifted delta function. The spectrum of a shifted delta function is a sinusoid.

Fast Fourier Transform (contd.)

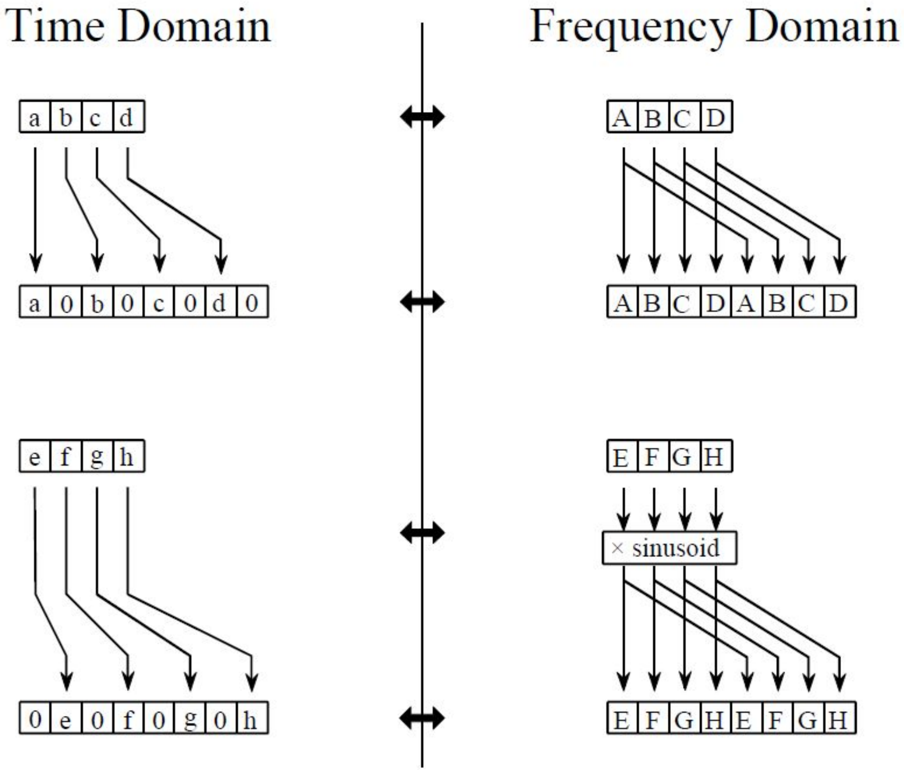
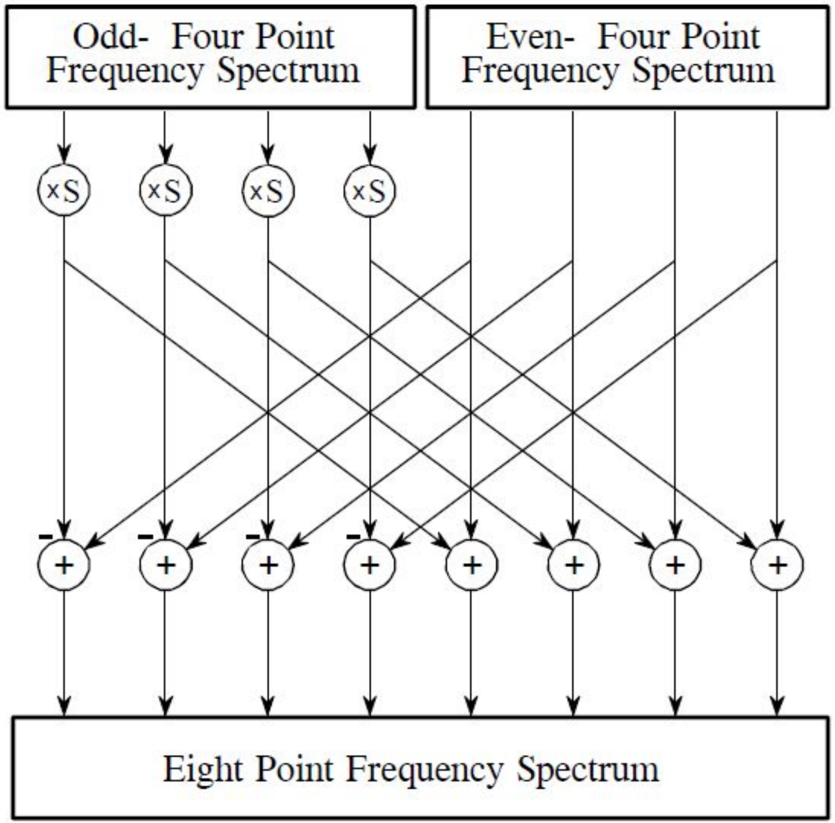


FIGURE 12-4
The FFT synthesis. When a time domain signal is diluted with zeros, the frequency domain is duplicated. If the time domain signal is also shifted by one sample during the dilution, the spectrum will additionally be multiplied by a sinusoid.

Fast Fourier Transform (contd.)

FIGURE 12-5
FFT synthesis flow diagram. This shows the method of combining two 4 point frequency spectra into a single 8 point frequency spectrum. The ×S operation means that the signal is multiplied by a sinusoid with an appropriately selected frequency.



Fast Fourier Transform (contd.)

FIGURE 12-6
The FFT butterfly. This is the basic calculation element in the FFT, taking two complex points and converting them into two other complex points.